SOFTWARE TESTING
PRACTICAL REPORT

PRACTICAL REVIEW — PR-14

Student Name	Parth Fegade
Enrollment Number	2504030100083
Programme	BCA — Semester 3
Practical	PR-14 — Selenium Web Automation: Login and Cart Validation
Tools	Python, Selenium WebDriver, Google Chrome, PyCharm, SauceDemo

1. Aim

To automate a shopping workflow on SauceDemo using Selenium WebDriver: open the website, enter valid credentials, validate login, add a product to the cart, open the cart, and verify the resulting URL.

2. Objectives

- Launch Chrome and navigate to SauceDemo.
- Enter username and password and submit the login form.
- Verify that the inventory page opens successfully.
- Add the Sauce Labs Backpack to the cart.
- Open the cart and validate the URL.
- Close the browser after execution.

3. Tools and Environment

Tool / Parameter	Purpose / Details
Python 3.x	Runs the automation script
Selenium WebDriver	Locates elements and automates browser actions
Google Chrome	Browser under test
PyCharm	IDE used to write and run the Python script
SauceDemo	Demo e-commerce website
Target URL	https://www.saucedemo.com/

4. Procedure

1. Create a Python project in PyCharm and create a file named pr14.py.

2. Install Selenium with pip install selenium if it is not already installed.

3. Import WebDriver, locator strategies, explicit waits, and expected conditions.

4. Launch Chrome, maximize the window, and open the SauceDemo login page.

5. Enter standard_user and secret_sauce, then click Login.

6. Wait for the inventory URL and assert that it contains inventory.

7. Click the Sauce Labs Backpack Add to cart button.

8. Click the shopping cart icon, wait for the cart URL, and assert that it contains cart.

9. Close the browser in a finally block to ensure cleanup.

5. Source Code

The code includes explicit waits and validation checks for login, product selection, and cart navigation.

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

driver = webdriver.Chrome()
driver.maximize_window()
wait = WebDriverWait(driver, 10)

try:
    # 1. Open SauceDemo
```

```
    driver.get("https://www.saucedemo.com/")
    print("Website opened")

    # 2. Enter valid login credentials
    wait.until(
        EC.visibility_of_element_located((By.ID, "user-name"))
    ).send_keys("standard_user")
    driver.find_element(By.ID, "password").send_keys("secret_sauce")

    # 3. Submit login form and validate inventory page
    driver.find_element(By.ID, "login-button").click()
    wait.until(EC.url_contains("inventory"))
    assert "inventory" in driver.current_url
    print("Login Validation: PASS")

    # 4. Add Sauce Labs Backpack to the cart
    wait.until(
        EC.element_to_be_clickable(
            (By.ID, "add-to-cart-sauce-labs-backpack")
        )
    ).click()
    print("Product Selection: PASS")

    # 5. Open cart and validate navigation
    driver.find_element(By.CLASS_NAME, "shopping_cart_link").click()
    wait.until(EC.url_contains("cart"))
    assert "cart" in driver.current_url
    print("Cart Validation: PASS")
    print("Tutorial 14 Completed Successfully")

finally:
    driver.quit()
```

6. Screenshot Evidence

Screenshots document the SauceDemo login page, product inventory, PyCharm source code, and execution output.

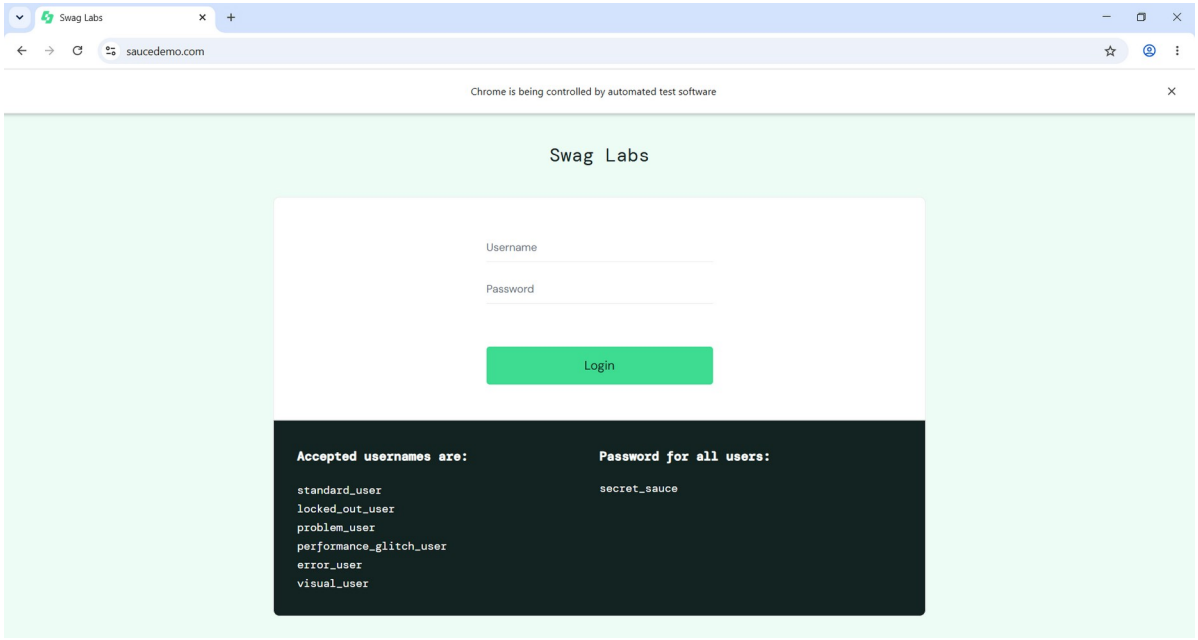


Figure 1. SauceDemo login page with accepted test credentials displayed on the page.

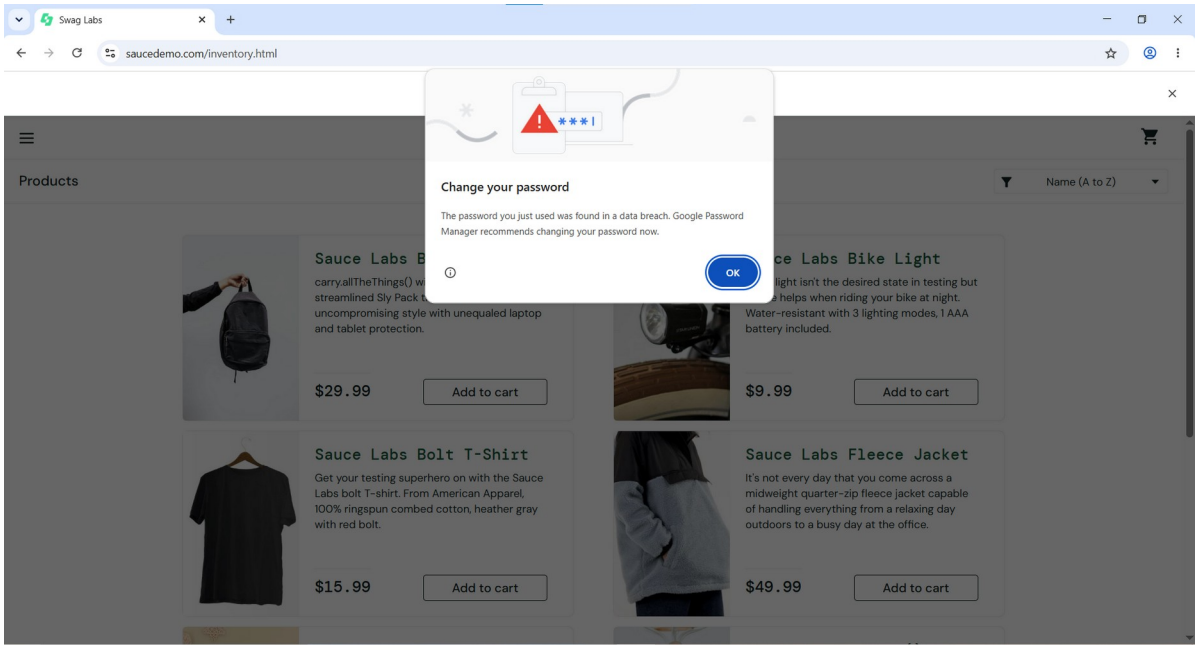


Figure 2. SauceDemo inventory page with product cards visible.

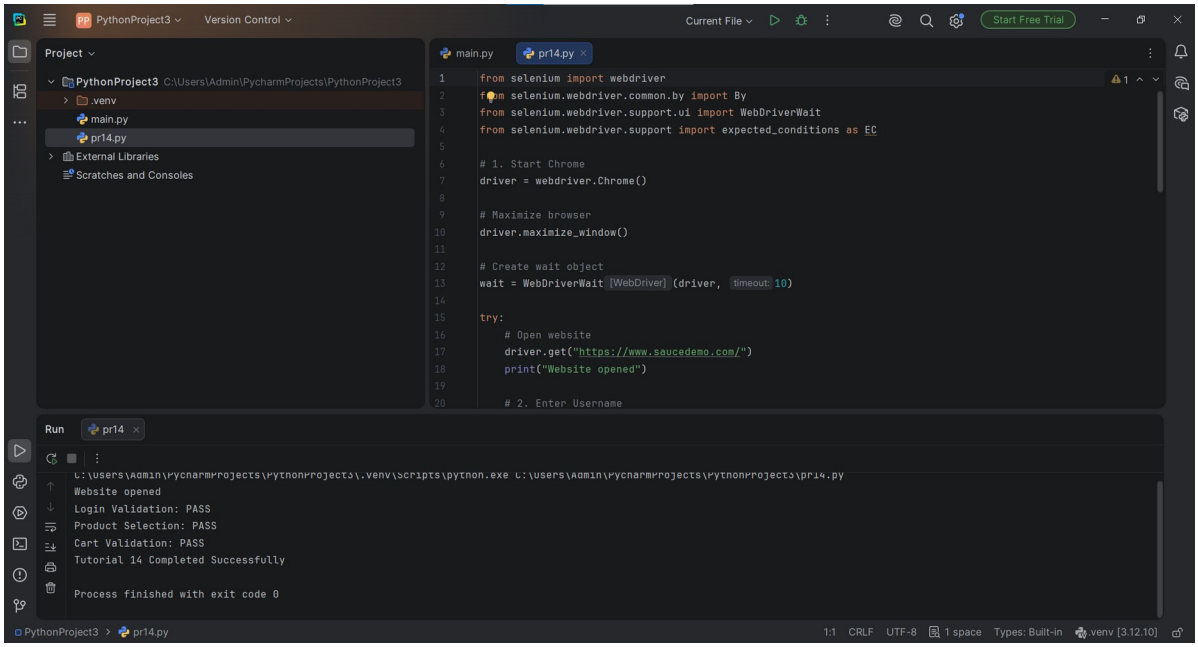


Figure 3. PyCharm source code for the Selenium automation script.

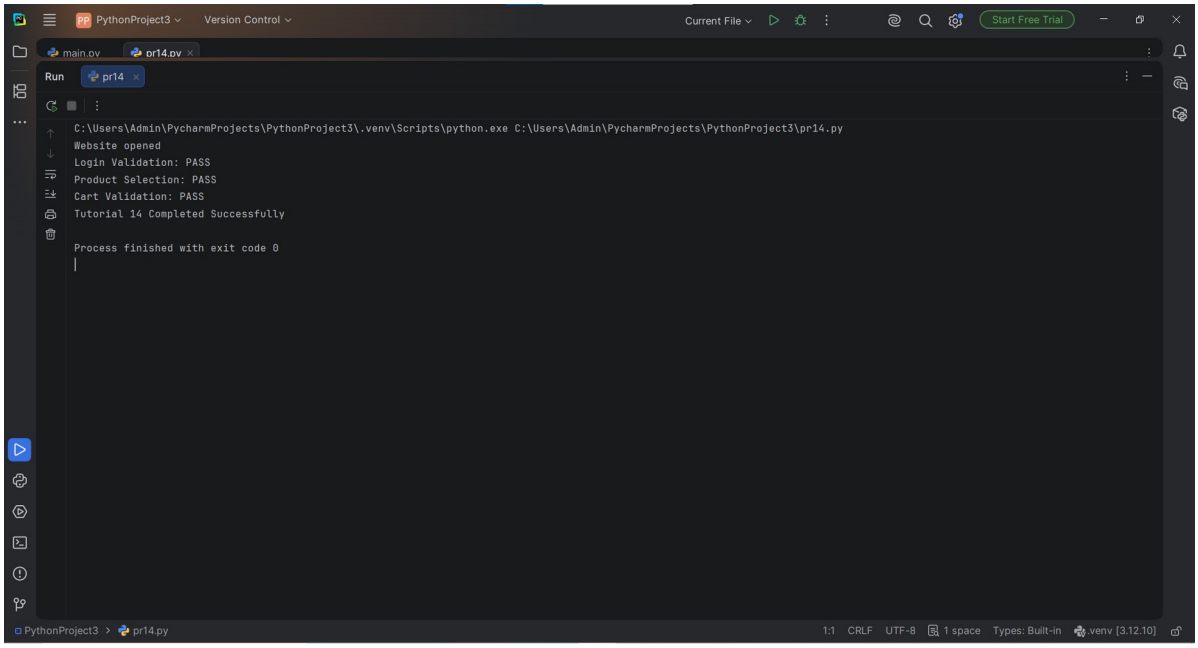


Figure 4. PyCharm console output showing login, product selection, and cart validation as PASS.

7. Test Case Summary

ID	Test	Expected Result	Status / Evidence
TC-01	Open website	SauceDemo login page loads	Pass — login screenshot
TC-02	Login	Valid credentials open inventory	Pass — inventory screenshot and console evidence
TC-03	Add product	Backpack is added to cart	Pass — console reports Product Selection: PASS
TC-04	Open cart	Cart page opens	Pass — console reports cart validation
TC-05	Validate URL	URL contains cart	Pass — assertion included

			in script

8. Observations

- Selenium opened the SauceDemo website in Chrome.
- The supplied console screenshot reports Login Validation, Product Selection, and Cart Validation as PASS.
- The browser screenshot shows the product inventory page; the source code includes cart navigation and URL assertions.

9. Result

The Selenium automation workflow was executed for SauceDemo login, product selection, and cart navigation. The supplied console output reports the main validation steps as PASS and the process exits successfully.

10. Conclusion

This practical demonstrates how Selenium WebDriver can automate browser navigation, handle login forms, interact with web elements, and validate application state using explicit waits and assertions.